

First Terminal Examination - 2080

Subject: Computer Science

Class : XI

(4271)

Full Marks: 50

Time: 2 hrs.

Set - A

Pass Marks: 20

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

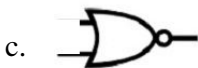
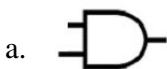
Group 'A'

Multiple choice Questions

[9 × 1 = 9]

Rewrite the best alternative of each question in your answer sheet.

1. Whenever the data is found in the cache memory it is called as
 - a. Cache Hit
 - b. Cache Miss
 - c. Cache Found
 - d. Cache Error
2. The speed of computer can be measured in term of
 - a. Megabits and Gigabits
 - b. Megabytes and Gigabytes
 - c. Megahertz and Gigahertz
 - d. All of the above
3. How to represent Boolean $F = (x + y)$ in logic gate?



4. One pico second means
 - a. 10^{-3} of seconds
 - b. 10^{-15} of seconds
 - c. 10^{-9} of seconds
 - d. 10^{-12} of seconds
5. What is gutter margin?
 - a. Margin that is added to the left margin when printing
 - b. Margin that is added to the right margin when printing
 - c. Margin that is added to the binding side of the page when printing

- d. Margin that is added to the outside of the page when printing
6. The binary equivalent of $(5.12)_8$ is
- a. 101.001101 b. 101.010110
c. 101.001010 d. 101.001100
7. 2's complement of 1011 is
- a. 0100 b. 0101 c. 0111 d. 1100
8. The NOR gate output will be high if the two inputs are
- a. 0 0 b. 0 1 c. 1 0 d. 1 1
9. $(E7)_{16} = (?)_8$
- a. 743 b. 447 c. 176 d. 347

Group 'B'

Give short answer to the following questions.

[5 × 5 = 25]

1. What is mobile computing? Explain the features of mobile computing.
2. Subtract $(111111)_2$ from $(1101)_2$ using 1's and 2's complement method.
3. Differentiate between primary memory and secondary memory.
4. What is word processor? Explain the features of word processor.
5. Convert following numbers as indicated in the bracket.
a) $(307.12)_8 = (?)_{16}$
b) $(123.625)_{10} = (?)_{16}$

Group 'C'

Give long answer to the following question.

[2 × 8 = 16]

6. Explain computer architecture with block diagram and functions of its components.
7. Prove the both associative law, distributive law and commutative law with their truth table and also draw graphical symbol.

*****Best of Luck*****

